



**Optimization Frameworks for Hybrid Biomass Power-to-Methanol System: MILP
and Reinforcement Learning Approaches**

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Abstract

Acknowledgment

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List of Abbreviations

1 Introduction

1.1 Background

The large-scale transformation of industrial energy systems is widely regarded as a critical enabler for achieving global decarbonization targets, particularly in so-called hard-to-abate sectors [1]. Among the available pathways, Power-to-X (PtX) technologies have emerged as a promising solution, enabling the conversion of renewable electricity into chemical energy carriers and synthetic fuels suitable for industrial applications [2]. However, PtX systems are inherently complex and capital-intensive, especially when their operation is coupled with the variability and intermittency of renewable energy source [3]. This complexity introduces significant challenges in both system design and operational planning, thereby necessitating robust optimization strategies that can effectively minimize capital expenditures (CAPEX) and operational expenditures (OPEX).

To address this need, Mixed-Integer Linear Programming (MILP) has become the prevailing industrial standard due to its robustness, transparency, and well-established solution techniques [4]. At the same time, the rapid advancement of artificial intelligence (AI) has motivated the exploration of alternative methodologies, particularly reinforcement learning (RL) [5]. In this context, RL can be understood as an agent-based framework that learns optimal operational strategies through iterative interaction with the system environment, effectively acting as an autonomous plant operator.

Despite its widespread adoption, MILP relies on linear approximations of system behavior, which can limit model fidelity when representing inherently nonlinear processes [6]. In practice, such nonlinearities are often simplified or neglected. This limitation makes RL particularly attractive, as it allows for flexible environment design in which high-fidelity, nonlinear system dynamics can be directly incorporated. Through appropriately defined reward structures, the RL agent can learn operational strategies that account for these complexities without requiring explicit linearization [7].

In summary, this thesis evaluates the techno-economic performance of a reinforcement learning–

based approach against a conventional MILP-based framework for a hybrid biomass power-to-methanol plant located in South Africa. The analysis is performed across multiple scenarios and case studies to capture a range of operating conditions and system configurations. Through this comparative assessment, the study seeks to clarify the role and current capabilities of RL in optimizing complex Power-to-X systems, while benchmarking its performance against an established industrial standard.

1.2 Introduction to Company

This master's thesis is conducted in cooperation with FEV Europe GmbH, specifically within its energy+resources department. In this industrial context, the research aligns with ongoing efforts to expand capabilities beyond traditional automotive engineering toward energy-related applications.

FEV Europe GmbH is widely recognized as an established expert within the automotive industry, particularly for its benchmarking and consulting services. Building on this foundation, the company is increasingly directing its efforts toward supporting innovation within the energy sector, reflecting broader industry trends and internal strategic priorities.

Within this framework, the present thesis contributes to the identification and evaluation of an efficient optimization tool for innovative energy solutions, with a focus on complex polygeneration plants. The outcomes are intended to support future FEV Europe GmbH projects by enabling improved system design and operational decision-making.

1.3 Objective

This thesis aims to systematically evaluate and compare two optimization frameworks across multiple operational scenarios. The analysis focuses on assessing their ability to represent and optimize complex hybrid energy systems, with particular attention to differences in system configuration and performance outcomes.

Building on this motivation, this thesis adopts a two-step methodological framework to systematically compare different levels of modeling fidelity and optimization approaches. In the

first step, a comparison is conducted between a fully integrated MILP formulation and a decoupled framework in which system design (sizing) and operation (dispatch) are optimized separately. This step aims to assess the impact of incorporating nonlinear CAPEX assumptions in the outer-loop (design optimization). In the second step, the resulting optimal system sizing is used as an input for operational optimization, where both MILP and RL-based approaches are applied in the inner-loop (dispatch optimization). This structure enables a direct comparison of the two frameworks in terms of their ability to determine cost-optimal operational strategies, particularly when contrasting linearized system representations with higher-fidelity nonlinear models.

A central objective is to investigate whether a non-linear physical model integrated with a reinforcement learning (RL) agent converges to substantially different component sizing and system layouts compared to a mixed-integer linear programming (MILP) approach. These differences are examined in terms of key techno-economic indicators, including capital expenditure (CAPEX), operational expenditure (OPEX), clipped energy, simulation time, amount of methanol produced and levelized cost of methanol (LCOM).

Based on the simulation results, the thesis further aims to identify and explain the underlying reasons for any observed discrepancies—or the lack thereof—between the two frameworks. This comparative analysis supports a final assessment of whether one framework demonstrates superior performance or whether its suitability is dependent on specific system configurations and operating conditions.

The expected contribution of this work is to enable the selection of an appropriate optimization framework for hybrid methanol synthesis plants, while also providing a structured approach for framework selection in other industrial energy systems. Additionally, the findings aim to improve optimization accuracy and/or reduce computational effort, and to enhance understanding of where and how RL-based approaches can be effectively applied within polygeneration system design.

1.4 Thesis Configuration

This thesis is structured to provide a systematic and transparent comparison of the selected optimization frameworks. Chapter 2 reviews the state-of-the-art literature on the application of

reinforcement learning (RL) in optimizing complex industrial processes and energy systems, including existing comparative studies between optimization approaches.

Chapter 3 details the methodological framework adopted for the comparison. It outlines a multi-stage evaluation procedure, describing each step taken to ensure a consistent and rigorous assessment of the two optimization approaches.

Chapter 4 presents and discusses the results of this multi-stage comparison. It analyzes the performance of each framework across different scenarios and case studies, and introduces a generalization scoring approach to assess their robustness and applicability across multiple evaluation dimensions.

Finally, Chapter 5 summarizes the main findings of the thesis and reflects on key limitations, assumptions, and methodological considerations that underpin the analysis, providing context for the interpretation and transferability of the results.

2 Literature Review

2.1 Hybrid Biomass Power-to-Methanol System

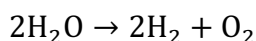
The hybridization strategy considered in this thesis integrates two upstream pathways that jointly supply the feedstock for methanol synthesis: (i) a biomass-derived pathway and (ii) a hydrogen pathway based on water electrolysis. This configuration enables coordinated carbon and hydrogen management prior to the synthesis loop. In the literature, gasification is most commonly employed to directly convert solid biomass into syngas. An alternative route relies on anaerobic digestion (AD), where biomass is first converted into biogas and then reformed to syngas prior to methanol synthesis. The differences in process steps primarily drive the trade-offs between these technologies. The AD-based route is attractive because it can be integrated with existing waste-treatment infrastructure; however, it requires additional downstream conversion steps, making the pathway less direct and more dependent on reforming, CO₂ handling, and potentially electrolysis integration. In contrast, gasifier-based routes are more established for methanol synthesis, as they provide syngas directly and are well suited for coupling with high-temperature processes. Their disadvantages include challenges related to high-temperature solid handling, the requirement for an oxygen supply (e.g., via an air separation unit, ASU), and stricter gas-cleaning requirements [8].

Table 1 Structural and Process Comparison between AD and Gasification routes

Metric	Anaerobic Digestion (AD) Route	Biomass Gasification Route	References
Primary Conversion Mechanism	Biochemical (mesophilic/thermophilic digestion of wet organic feedstocks)	Thermochemical high-temperature reactions (> 800°C – 1200°C)	[9]
Intermediate Stream	Biogas (~ 55--65% CH ₄ , 35--45% CO ₂) requiring reforming to form syngas.	Raw syngas (CO, H ₂ , CO ₂ , CH ₄) containing tars and particulates.	[9], [10]
Feedstock Flexibility	High-moisture, biodegradable substrates (e.g., sewage sludge, organic municipal solid waste, agricultural waste,	Low-moisture solid biomass (e.g., forestry residues, lignocellulosic crop waste, refuse-derived fuel/RDF).	[9]

	slurry).			
Core Process	Biogas reforming (Dry Severe syngas conditioning [9], [11]			
Bottleneck	Reforming DRM / Steam requirements (tar removal, Reforming SMR) and particulate scrubbing, stoichiometric H ₂ /CO ratio sulfur/halogen cleanup, and adjustment. Water-Gas Shift).			
Techno-Economic Profile	Lower CAPEX per unit capacity; economically viable at smaller, regional plant sizes.	High upfront CAPEX; strongly dependent on economies of scale and centralized feedstock supply chains.	[10]	
Lifecycle GHG Abatement	Moderate-to-high carbon footprint reduction depending on feedstock cultivation and digestate handling.	Exceptional carbon abatement potential (up to −0.88 t CO ₂ e/t CH ₃ OH) when combined with biogenic carbon capture.	[10]	

The hydrogen pathway begins with the electrochemical splitting of water into hydrogen and oxygen using an electrolyzer.



Equation 1 Electrolysis.

Several studies identify Solid Oxide Electrolyzer Cells (SOECs) as a promising option for power-to-methanol systems. The high operating temperature of SOECs allows for heat integration and potential oxygen utilization, which can increase overall plant efficiency compared to low-temperature systems[12]. However, key challenges include long-term stack durability, technological maturity, and limited operational flexibility, which is particularly relevant for systems coupled to variable renewable electricity. Techno-economic comparisons suggest that alkaline water electrolysis (AWE) represents a viable short- to mid-term option, whereas SOECs are more likely to be competitive in the mid- to long-term. Low-temperature electrolyzers such as AWE and proton exchange membrane (PEM) systems offer higher technological maturity and superior load-following capability, which is advantageous for renewable integration, but they do not fully exploit heat integration or co-electrolysis opportunities available to SOECs [13]. A

detailed quantitative synthesis of these performance metrics and operating limits is summarized in Table 2.

Table 2 Parameter comparison between AWE, PEM, and SOEC [14], [15], [16].

Parameter		Alkaline Electrolysis (AWE)	Water Proton Exchange Membrane (PEM)	Solid Electrolysis (SOEC)	Oxide
Operating Temperature (°C)		60 – 90°C	50 – 80°C	650 – 1000°C	
Operating Pressure (bar)		< 30 bar	30 – 80 bar	< 30 bar	
Energy Consumption (kWh/Nm ³ H ₂)		4.5 – 5.5	4.2 – 5.5	3.2 – 3.8	
Minimum Operational Load (% <i>P_{nom}</i>)		10% – 20%	0% – 10%	20% – 30%	
Dynamic Response		Moderate (0.2--20%/s)	Very (10--100%/s)	High	Low (< 1%/s)
H ₂ Purity (%)		> 99.5%	> 99.999%	> 99.9%	
Commercial Status / TRL		Commercial / State-of-the-art (TRL 9)	Near-commercial (TRL 8-9)	R&D (TRL 6-7)	

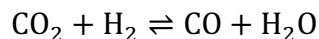
In hybrid biomass and power-to-methanol (PBtM) systems, the technical consensus across recent literature emphasizes the need to achieve an optimal blending between the carbon-rich syngas with the pure hydrogen coming from the electrolyzer to achieve complete carbon efficiency [17]. Three repeating strategies are used in the literature for this specific goal:

1. Direct hydrogen blending upstream of methanol loop compressor:

Biogas goes through a reformer to produce a primary syngas containing CO, CO₂, and H₂. Due to the low stoichiometric module below the required threshold of around 1.5 – 1.8, the blending with hydrogen before the methanol reactor increases this number to >2, enabling 100% biogenic carbon recovery without venting CO₂ [18].

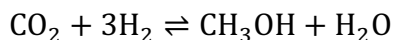
2. Reverse Water-Gas Shift (RWGS) Coupling Junction:

Unrefined or partially reformed biogas CO₂ is routed into an RWGS reactor



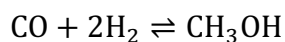
Equation 2 RWGS chemical reaction.

Where it explicitly mixes with a portion of the green hydrogen stream at 600 – 800°C. Direct CO₂ hydrogenation to methanol produces one mole of water for every mole of methanol,



Equation 3 CO₂- based Methanol synthesis.

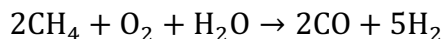
which accelerates catalyst deactivation through kinetic inhibition and active site sintering [17]. By meeting in an RWGS unit prior to the methanol synthesis, the hydrogen reduces CO₂ to CO, shifting the synthesis reaction toward the faster, more thermodynamically favorable CO hydrogenation route [19].



Equation 4 CO-based Methanol synthesis.

3. Anodic Oxygen Integration at the Reforming Stage:

In this strategy, the byproduct of the electrolysis step, O₂, already meets the biogas path inside an Autothermal Reformer (ATR). The oxygen from the electrolyzer will then be used to generate internal thermal energy via exothermic methane partial oxidation



Equation 5 ATR chemical reaction.

Which sustains the endothermic steam reforming reactions. This strategy eliminates the need for a capital-intensive Air Separation Unit (ASU). Dossow et al. [20] demonstrated that full oxygen integration between water electrolysis and thermochemical reforming reduces overall auxiliary plant electrical consumption by 8-12%, establishing a closed mass-and-energy synergy between the hydrogen and biogas paths.

While these strategies establish the thermodynamic and chemical validity of stream integration, maintaining mass-balance ratios between the continuous biogas path and volatile power-dependent hydrogen path remains a central operational dispatch challenge requiring advanced optimization strategies [21].

The hybrid approach to methanol production is attractive due to its potential for higher carbon utilization and improved use of biomass resources. One study reports a carbon utilization of 89.8% and an overall efficiency of 85.8%, with up to 101% higher methanol production per unit of biomass compared to a conventional biomass-to-methanol baseline [22]. Another analysis indicates that a state-of-the-art biomass-to-methanol process achieves approximately 53.3% efficiency with a 4.8-year payback period, while integrating electrolysis for full carbon conversion increases efficiency to 64.5–66.0%, albeit with less favorable economics [23]. Furthermore, under favorable electricity price conditions, flexible operation can allow the plant to capture value from temporal price variations instead of operating continuously at a fixed load [24].

Despite these advantages, several bottlenecks challenge the feasibility of hybrid methanol systems. These include electrolyzer utilization rates, electricity price volatility, capital cost and lifetime of electrolyzer stacks, as well as the integration of oxygen and heat streams. For example, it has been reported that high electrolyzer capacity factors are required for competitive e-methanol production, and that full-carbon-conversion configurations become attractive only when improvements in electricity costs, stack lifetime, and capital expenditure are realized [25]. Oxygen management represents an additional practical constraint: while direct integration of oxygen streams may reduce the need for large-scale oxygen storage and improve capital efficiency and safety, this is contingent on matching oxygen supply and demand within the plant [21].

To address these challenges, process and system-level optimization emerges as a critical tool for improving both technical and economic performance. For instance, optimization of the methanol synthesis reactor in a power-and-biomass-to-methanol (PBtM) configuration has been shown to reduce e-methanol production costs by approximately 20% [25]. While such studies typically focus on individual subsystems, it is reasonable to expect that plant-wide optimization could yield further cost reductions and efficiency gains. In this context, optimization acts as the bridge between technically feasible system designs and configurations that remain viable under realistic operating conditions, including fluctuating electricity prices, electrolyzer degradation, and large-scale capital constraints.

2.2 MILP for Complex Polygeneration System

MILP, or Mixed-Integer Linear Programming, has been used as a tool to optimize different kinds of energy or industrial plants by linearizing the model of these processes and setting up a linear objective function to either maximize or minimize specific key performance indices (KPIs) based on the optimization objective [26]. In an energy system that includes the flow of electricity, heat, and mass, the constraints considered will then dictate how the balance of the system should behave, the minimum and maximum operational dispatch, and the evolution of the storage state of charge, if storage exists. On the other hand, the objective function mainly includes the capital expenditure (CAPEX) and operational expenditure (OPEX) of the models defined in the system.

$$\begin{aligned} \min Z = & \sum_{c \in C} \left[\text{CAPEX}_c \cdot \left(\frac{r(1+r)^{n_c}}{(1+r)^{n_c} - 1} \right) + \text{OPEX}_{\text{fix},c} \right] \\ & + \sum_{t=1}^T \left(C_{\text{electricity}}(t) + C_{\text{feedstock}}(t) + C_{\text{var}}(t) \right) \cdot \Delta t \end{aligned}$$

Equation 6 Objective function example of an energy system

$$\begin{aligned} \sum_{g \in G} P_g(t) + P_{\text{grid, buy}}(t) + P_{\text{bat, dis}}(t) &= P_{\text{demand}}(t) + P_{\text{grid, sell}}(t) + P_{\text{bat, ch}}(t) + \sum_{c \in C} P_c(t) \quad \forall t \\ &\in T \end{aligned}$$

Equation 7 Power balance constraint example

$$P_{\min,c} \cdot y_c(t) \leq P_c(t) \leq P_{\max,c} \cdot y_c(t), \quad y_c(t) \in \{0,1\} \quad \forall c \in C, \forall t \in T$$

Equation 8 Example of a flow constraint

$$\text{SoC}(t) = \text{SoC}(t-1) \cdot (1 - \sigma) + \left(\eta_{\text{ch}} \cdot P_{\text{bat, ch}}(t) - \frac{P_{\text{bat, dis}}(t)}{\eta_{\text{dis}}} \right) \cdot \Delta t \quad \forall t \in T$$

Equation 9 Example of SOC evolution definition in MILP.

Together, the MILP will generate a global equation that accounts for the constraints and the objective function, and, using the MILP solver, it will perform relaxation and branch-and-bound to find the valleys or hills within the linear boundaries. In practice, MILP has been used as the reference method for operational optimization, demand response, scheduling in energy communities, industrial plants, and integrated multi-energy systems because it produces globally optimal solutions to the defined linear problem when sufficient computation time is available [27].

Selecting an appropriate optimization methodology for complex polygeneration systems requires balancing physical model accuracy, computational tractability, and handling of operational uncertainty [28]. Adjacent to Mixed-Integer Linear Programming (MILP), several mathematical and heuristic paradigms are widely evaluated across the literature [29]. Mixed-Integer Non-Linear Programming (MINLP) is employed when non-linear thermo-chemical physics are prevalent; however, as the system scale expands, non-convexities can cause computational complexity to grow exponentially, rendering large-scale models intractable or prone to local sub-optima [30]. Stochastic programming explicitly addresses parameter uncertainty by evaluating system states across multi-scenario trees, but its computational burden rises sharply with increasing scenario counts and temporal horizons [31]. For non-linear, multi-objective hybrid energy problems, metaheuristics such as Particle Swarm Optimization (PSO) and Genetic Algorithms (GA) are frequently adopted due to their model-free flexibility and robustness; yet, unlike exact MILP solvers, they offer no mathematical guarantee of global optimality [32]. To bridge these trade-offs, hierarchical and bilevel optimization architectures decouple strategic capacity sizing (e.g., using metaheuristics or Bayesian optimization) from high-frequency operational dispatch (e.g., solved via MILP or Reinforcement Learning), a strategy detailed in subsequent sub-sections [33].

Table 3 Advantages and Disadvantages of different optimization paradigms [32], [33], [34], [35].

Optimization Paradigm	Key Advantages	Key Disadvantages
MILP <i>(Mixed-Integer Linear Programming)</i>	• Proven global optimality. • Ultra-fast commercial solvers (Gurobi/CPLEX). • High bankability & industrial standard.	• Cannot natively handle non-linear physics or exponential cost curves without piecewise linear approximations.
MINLP <i>(Mixed-Integer Non-Linear Programming)</i>	• Captures highly realistic physical/chemical behavior without linear simplifications.	• Non-convex problems can get trapped in local sub-optima. • High computational complexity (NP-hard).
Stochastic Programming	• Explicitly accounts for parameter uncertainty (e.g., volatile weather, price	• Suffers from scenario explosion; problem size scales exponentially with more

		shocks).	scenarios.
Dynamic Programming (DP)		• Solves complex sequential dynamic problems globally without needing convexity.	• Suffers from the "Curse of Dimensionality" as the number of state variables increases.
Metaheuristics (Genetic Algorithm, Particle Swarm Optimization, Bayesian Optimization)		• Model-free; handles arbitrary complex, black-box, or derivative-free environments easily.	• No mathematical guarantee of global optimality. • Can be slow to converge and requires extensive parameter tuning.
Decomposition / Bilevel (Benders, Stackelberg)		• Drastically reduces problem scale. • Enables solving complex non-linear sizing/dispatch problems independently.	• Convergence depends heavily on algorithmic cuts/tuning. • Formulations can be difficult to implement.

MILP has been used a lot of times in the literature to optimize different kinds of polygeneration systems. A similar study that researches the use of MILP to optimize a similar system of a hybrid biomass power-to-methanol system showed that MILP discovered optimal sizing and dispatch that results in LCOM of 805 EUR /t under the assumption of high biogas availability. This system adopts a configuration consisting of a low-pressure AWE, absorption-based CO₂ capture, and steam production from methanol purge gas for internal heat supply [36]. In one energy-community case, MILP has been used to jointly optimize design and operation of distributed polygeneration with district heating and cooling, storage, and energy sharing, showing clear cost-emissions tradeoffs. The results showed two outcomes can come out depending on the objective function emphasis of either a 24.3% CO₂ emission cut at a 1.9% cost increase or a 31.9% CO₂ cut with a 2.22.2% CO₂ emissions increase [37].

The key strength of MILP is the deterministic model that it offers on the modeled problem. If the system can be expressed with linear relationships, MILP guarantees an optimal solution to that formulation, which makes it highly attractive for sizing and dispatching problems with hard operational constraints [38]. Other than that, every constraint, feasibility, and infeasibility is

explicit and can be traced to how the system is defined; thus, it has the benefit of easier interpretability and debugging. Due to the exactness and deterministic model, MILP becomes a great baseline for benchmarking novel methodologies such as RL [39].

The main gap of MILP is that optimality is guaranteed only for the approximated linear model, not necessarily the real physical system [40]. Therefore, in the case of complex polygeneration systems that contain nonlinear part-load efficiencies, thermal inertia, degradation, recycle interactions, and stochastic renewables, they often need to be simplified to their linear equivalent or linearized through processes such as piecewise linearization [41]. Another significant drawback is also the computation time that significantly increases with the number of non-convex definitions, decision variables, and long horizon, which is the exact opposite to a trained RL agent that can run in milliseconds after being trained, and results in a major limitation for real-time operation [42].

The reason why RL becomes interesting to explore is not that MILP is weak at optimization, rather that it is less suitable for fast repeated control when the environment is uncertain, partially observed, and dynamically changing [43], [44]. In biomass power-to-methanol systems, this would matter when biomass quality, renewable power input, hydrogen availability, market signals, and flexible downstream demand vary over time, forcing frequent re-optimization under nonlinear couplings. In this niche, RL literature positions itself as a complementary method rather than a direct replacement for MILP, and MILP remains a robust baseline to validate the results of this RL approach [45].

2.3 State-of-the-art RL in Optimizing Energy and Industrial Systems

2.3.1 Reinforcement Learning Fundamentals

The adoption of RL to optimize an energy system or industrial process has been of interest to a lot of researchers due to its compelling advantages, particularly due to its inherent flexibility and adaptability [46]. In comparison with MILP, RL is different because it acts as a sequential decision problem, which formulates control policies directly from interaction data rather than relying on a complete mathematical model [47]. The consensus among scholars concludes that RL is especially attractive where renewable variability, stochastic demand, and sector coupling make exact model-based control difficult [48]. Thus, a hybrid biomass power-to-methanol system is a great subject

of analysis for the RL agent as it embodies all the characteristics mentioned.

By directly interacting with the system it is thrown into, the RL agent can derive optimal strategies through a series of Markov Decision Processes (MDPs). An MDP is described as a tuple $\{S, A, T, R\}$, where S denotes a set of states, A stands for a set of actions, T represents the transitional probability, and R denotes the reward function. This process involves a dynamic evolution of the states, actions, transitional probabilities, and reward at each time step, where the RL agent selects and executes an action at a current state and receives a reward from its action.

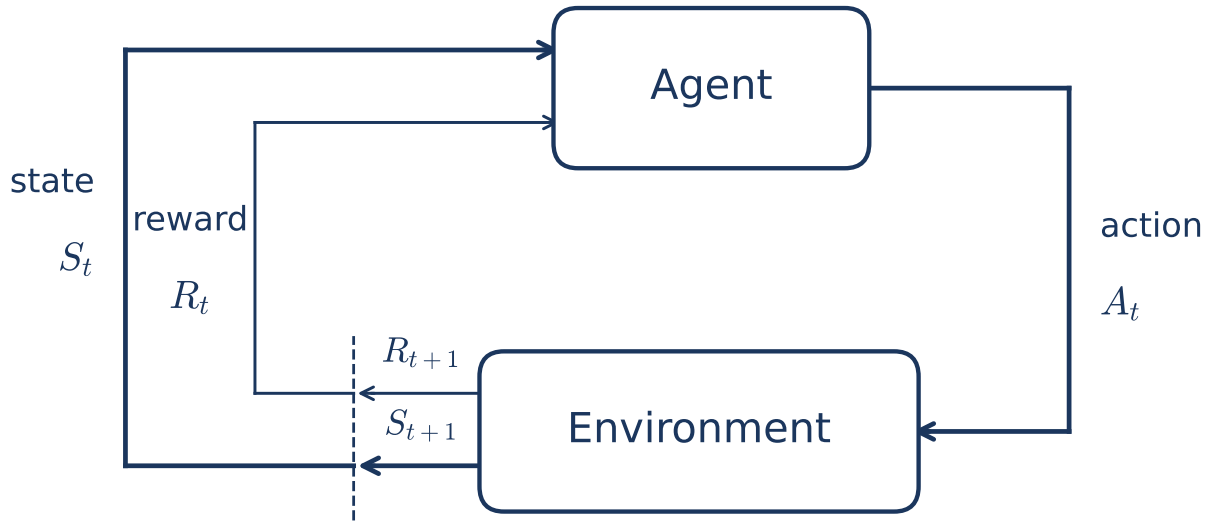


Figure 1 MDP process diagram.

The main objective of an agent is to maximize the discounted sum of rewards from time t to infinity, denoted by R_t , which is expressed as follows

$$R_t = R_{\{t+1\}} + \gamma R_{\{t+2\}} + \gamma^2 R_{\{t+3\}} + \dots = \sum_{k=0}^{\infty} \gamma^k r_{t+k+1}$$

Equation 10 Reward function of MDP process.

Where γ is a discount factor, specifying the future rewards. The agent's actions are guided by a policy π , which specifies what to do given the states of the environment. Each policy is associated with a state-value function $V^\pi(s)$, which predicts the outcome for state s when following policy

π :

$$V^\pi(s) = E[R_t | S_t = s]$$

Equation 11 State-value function.

Where E indicates the expected value. The optimal value of state s , $V^*(s)$, is defined to map the maximum value over all possible policies:

$$V^*(s) = \max_{\pi} E \left[\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t) \mid s_0 = s, a_t = \pi(s_t) \right]$$

Equation 12 Optimal state-value function.

The other related variable is an action-value function $Q^\pi(s, a)$, which gives the expected return when taking action a in state s and following policy π :

$$Q^\pi(s, a) = E[R_t | s_t = s, a_t = a]$$

Equation 13 Action-value function.

The optimal Q-value of a state-action pair, $Q^*(s, a)$ is the maximum Q-value over all possible policies:

$$Q^*(s, a) = \max_{\pi} E \left[\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t) \mid s_0 = s, a_0 = a, a_{t>0} = \pi(s_t) \right]$$

Equation 14 Optimal action value function.

In short, an optimal policy $\pi^*(s)$ is a policy whose state-value function is equal to the optimal value of the state. Modelling an accurate environment's dynamics through MDPs is emphasized by many researchers in the field to obtain a robust framework and good results [49]. Adopting this process enables RL agents to adapt to complex and dynamic environments effectively, which is ideal for optimizing the operation of a polygeneration plant [50].

To select an appropriate RL strategy for optimizing an energy system, it is paramount to understand the different structural arrangement that differentiates modern RL algorithms. RL algorithms are primarily categorized along two fundamental axes: (1) Model-based vs. Model-free RL, which specifies how the agent-environment dynamics, and (2) Value-Based, Policy-Based, and Hybrid Actor-Critic architectures, which determines how control policies are learned and updated. Depending on the type of system, data availability, and objective, one arrangement can be favored over the others.

2.3.2 Environmental Dynamics of RL: Model-Based vs. Model-free

- **Model-Based RL:** The agent uses an explicit model of the system dynamics to plan future actions by simulating potential trajectories (e.g. Dyna-Q, Monte Carlo Tree Search, Model Predictive Path Integral control). In other words, the knowledge of the model directly influences the agent's decision in choosing its next action. Although this approach introduces better sample efficiency, model-based RL often leads to longer online execution time and model-bias error which can be detrimental for highly complex and non-linear polygeneration system [1].
- **Model-Free RL:** The agent treats the model of the system as a black box and purely relies on the navigating the accessible levers and system feedback to dictate its policy. As a consequence, this strategy requires significantly more data and longer training time for the agent to finally understand the underlying physics of the system. However, with a robust and accurately representing digital twin of the system, the agent can use the digital environment to generate as much data as it needs during the training period to finally reach a good policy while avoiding model-bias because all the physics are governed inside the digital twin black box [1].

2.3.3 Policy Optimization Algorithms: Value-Based, Policy-Based, and Actor-Critic

There are different types of RL algorithms to be chosen for complex energy system optimization and control. The types of RL algorithms are usually divided into three parts: value-based RL, policy-based RL, and actor-critic, which sits in between them. The difference between these different types lies in the learning methodology for the RL agent. Value-based RL learns a value function, usually a $Q(s,a)$ estimate, and then chooses an action that returns the best estimated value. Some examples of value based RL are Q-learning and Deep Q Network (DQN) [51]. A study argues that value-function methods can run into convergence trouble, especially when the system is in continuous, multivariate settings. This algorithm excels in controlling and optimizing discrete choices but not continuous ones [52].

On the other hand, Policy-based RL learns the Policy directly, which means it parametrizes the action rule itself and pinpoints the optimal policy with methods like policy gradients and PPO [53]. There is much literature on RL that simply applies a value-based approach, however, since energy

problem are often continuous, high-dimensional, and constrained, policy-based approaches are usually preferred [54].

Actor-critic methods are a mix of the two: the actor learns the policy, while the critic learns the estimated value to make training more stable. In the literature, earlier studies still adopted Q-learning, but newer research starts to implement actor-critic and other deep RL variants for complex control tasks [55]. This actor-critic method includes Proximal Policy Optimization (PPO), Soft Actor-Critic (SAC), Asynchronous Advantage Actor-Critic (A3C), Twin Delayed Deep Deterministic Policy Gradient (TD3), and Deep Deterministic Policy Gradient (DDPG). Numerous studies have shown the application of this algorithm method into different kinds of energy systems and concluded that it performs best when uncertainty, continuous control, and multi-timescale coordination matter [47], [56].

2.4 RL vs MILP for Different Energy System Applications

In recent years, there have been multiple attempts in the literature to benchmark the results of RL and MILP for various energy system problems. Vetter *et al.* [26] analyzed the implementation of a deep reinforcement learning framework using the Deep Q-Network (DQN) algorithm to optimize the control of heat pump and heating element operation in a Grid-PV-Thermal Energy Storage (TES) system for an energy community. In this study, both frameworks are used to optimize only the dispatch strategy of the energy system in a fixed sizing environment. Furthermore, the RL agent's action is only limited to controlling the SOC of the thermal energy storage, while observing its impact on the environment or state it can see. On the other hand, the MILP framework decides the flow of grid power and heating element at each time step. The result of this research shows that both frameworks successfully produce yearly cost savings compared to the reference scenario, with MILP having greater cost savings of a 10.06% reduction compared to 8.78% for RL. Aside from the cost result, the RL solution leads in terms of other metrics such as CO₂ emission reduction, self-consumption ratio, and self-sufficiency ratio, without explicitly optimizing them. This is because RL decides to use more PV, which further reduces CO₂ emissions.

Other than benchmarking the two frameworks for operational dispatch, some researchers also conducted a comparative analysis on RL and MILP for optimizing both the design and control

decisions. Cauz *et al.* [57] used Battery-PV systems as their subject of analysis in order to test and optimize the performance of RL and MILP. The RL algorithm used here is the Direct Environment and Policy Search (DEPS). The design problem here lies in sizing the battery and PV power, whereas the control problem lies in dispatching battery and grid operations. The comparison is done in two stages: the first stage only involves one week of operation, and the second stage tests both frameworks' performance in a one-year case study. For the first stage, the authors discovered that under a fixed sizing scenario, both frameworks manage to converge to the global optimum. However, while introducing the design decision problem, the RL at first managed to produce a better result in terms of cost; however, the authors later found that this is due to the modelling limitations that the RL excludes, which is the battery SOC balance at the end that should be the same as the initial SOC state. With this assumption being included in the calculation, the cost result of RL drops, and MILP prevails to become cheaper. This shows how important the modelling phase and identical assumptions are for a fair comparison. Interestingly, the RL approach manages to converge to a near-optimal solution with a design variable that is significantly different from the MILP: 62 kWh and 41 kWp for RL and 40 kWh and 103 kWp for MILP for battery and PV, respectively, with the RL layout having 145.65% of the cost of MILP after considering the additional SOC calculations.

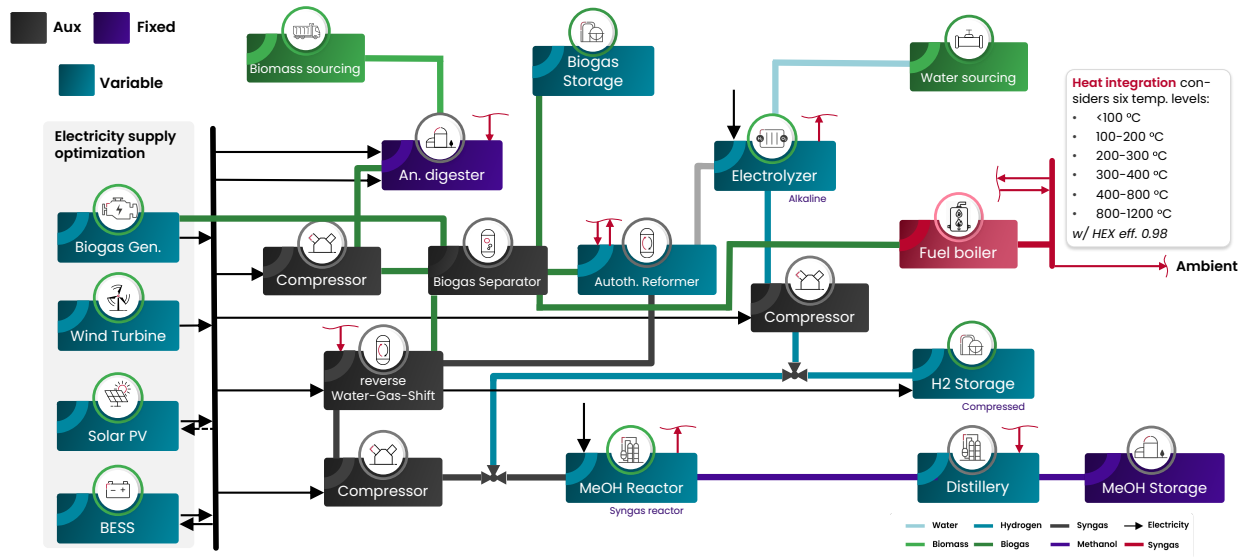
In the second stage of this study, the RL approach failed to converge to the same cost value of MILP under the fixed sizing scenario, with it being 12% more expensive. Under the joint operational and sizing decision optimization scenario, RL ends up having 22% more cost than MILP, while the design decisions are significantly smaller than MILP (44 kWh & 57 kWp for RL and 95 kWh & 81 kWp for MILP for battery and PV, respectively). According to the authors of the study, this poorer performance of RL could be due to a lot of reasons; two of the notable reasons include hyperparameter tuning and the outdated RL algorithm chosen, which was developed in 1992.

Reviews now identify TD3, SAC, Multi-Agent Reinforcement Learning (MARL), and PPO as the main advanced RL families being adopted for energy management, especially in online or uncertainty-heavy settings [58]. As an on-policy policy-gradient framework, PPO utilizes a clipped surrogate objective function that constrains policy updates within a selected trust region, which

ensures high training stability and steady convergence across continuous control domains [59]. In one study on optimizing the operation of multireservoir hydropower, Castro-Freibott et al. [60] implemented different RL algorithms (A2C, PPO, and SAC) in comparison to an MILP baseline. The experiment demonstrated that MILP achieves superior performance in the smaller system, but PPO was the strongest RL method tested and scaled far better than MILP when the system gets bigger. This result also shows that PPO could maintain consistent performance across discrete and continuous action spaces while achieving it with less than one second of execution time. Another study compared four RL algorithms: PPO, A2C, SAC, TD3 for a battery power plant control problem and demonstrated TD3 superior results among the RL algorithms and outperformed MILP by 5% in cost savings [61]. In a study on optimizing renewables penetration in modern smart grids, a comparative analysis was conducted between rule-based, MILP, and DRL algorithms (DDPG, PPO, SAC). The result demonstrated SAC superiority in delivering the best results, reducing operational costs by 20%, achieving 92.8 % renewable application, while minimizing loss-of-load probability [62]. From diving deep into the literature, it is possible to argue that many RL strategies are already competitive with MILP, or even some better; however, this is highly dependent on the system complexity and uncertainty assumption.

3 Methodology

3.1 Hybrid Biomass Power-to-Methanol System Model



The model developed for this comparative analysis represents a hybrid biomass power-to-methanol (PBtM) architecture. The complete high-level balance of plant (BOP) for this integrated system is illustrated in the preceding figure. Within this configuration, methanol production is driven by two parallel upstream conversion pathways: the generation of a carbon-rich stream via biomass anaerobic digestion, and the production of hydrogen utilizing an alkaline water electrolyzer (AWE).

Before reaching the methanol synthesis loop, the biogenic gas undergoes a sequence of intermediate conversion processes. Initially, raw biogas is fractionated into methane and carbon dioxide; the methane is routed to an autothermal reformer (ATR), while the carbon dioxide is directed to a reverse water-gas shift (RWGS) reactor. Within the ATR, the methane undergoes partial oxidation using byproduct oxygen supplied by the AWE. This triggers an autothermal reforming process, generating a primary syngas rich in carbon monoxide (CO) and hydrogen.

This intermediate syngas is subsequently fed into the RWGS unit, where it mixes with the

previously separated carbon dioxide. The RWGS reactor serves a critical regulatory function, converting carbon dioxide into additional CO to dynamically balance the hydrogen-to-CO ratio required for optimal methanol synthesis, thereby introducing operational flexibility depending on real-time limiting factors. In parallel, the AWE generates ultra-pure hydrogen, which is subjected to a multi-stage compression process before converging with the conditioned syngas immediately upstream of the methanol synthesis reactor. The combined upstream gases are then catalytically converted into crude methanol, which undergoes distillation to achieve the required purity prior to transport and storage.

For both the biogenic and hydrogen paths, an intermediary storage of Biogas Storage (BGSS) and Compressed Hydrogen Gas Storage (CGH₂) are introduced. These storage components allow the system to react to the variable renewable energies

The system's electrical demand is satisfied by on-site renewable energy installations, comprising solar photovoltaic (PV) and wind turbine capacities. To ensure operational reliability and mitigate the intermittency of these renewable sources, a backup biogas generator is integrated into the architecture; this generator is fueled by methane routed directly from the upstream biogas separation unit.

For the biogenic, hydrogen, and electricity paths, an intermediary storage of Biogas Storage (BGSS), Compressed Hydrogen Gas Storage (CGH₂), Battery Energy Storage System (BESS) have been introduced. These storage components allow the system to react to the variable renewable energies to find the optimal dispatch in maximizing methanol production. However, this is subject to the solver decision as the solver has the ability to decide the economic competitiveness of these components in reducing total cost or LCOM.

Conversely, the primary thermal energy requirements are provided by a high-temperature fuel boiler operating within the 800–1200 °C range. To maximize overall thermodynamic efficiency, the thermal management system employs a cascaded heat integration strategy. As dictated by the system architecture, this heat recovery network is discretized into six distinct temperature levels ranging from below 100 °C up to 1200 °C, to systematically capture and recycle waste heat from

the various exothermic conversion processes occurring throughout the plant.

To formulate a rigorous computational optimization problem, the system components are mathematically classified into three fundamental categories based on their sizing logic and operational constraints:

- **Fixed Variables:** These parameters define the pre-determined infrastructural baseline required for the plant's operation. By enforcing mandatory capital investments for these core components, a hard constraint is imposed on the solver. This crucially prevents the optimization algorithm from trivially converging on a zero-capacity, zero-cost null state.
- **Decision Variables:** These represent the primary control parameters that the optimization methodology actively solves. The algorithm iteratively determines the optimal installed capacities for these components to minimize overall system costs, operating strictly within the boundaries established by the fixed variables.
- **Auxiliary Variables:** The sizing of these dependent, balance-of-plant components is not determined independently by the solver. Instead, their capacities are deterministically scaled as a direct mathematical function of the corresponding fixed and decision variables.

The subsequent table summarizes the scaling logic for the system architecture, detailing how each component's capacity is dimensioned relative to its designated lead variables.

Components	Types	Lead Variables	Dependency
AD	Fixed	Biogas	-
MeOH Storage	Fixed	Methanol	-
WT	Variable	Electricity	-
PV	Variable	Electricity	-
BESS	Variable	Electricity	-
AWE	Variable	Electricity	-
ATR	Variable	Syngas	-
CGH2	Variable	Hydrogen	-

BGSS	Variable	Biogas	-
Generator	Variable	Electricity	-
MeOH reactor	Variable	Methanol	-
Distillery	Variable	Methanol	-
Biogas Compressor	AUX	Electricity	AD
Steam Generator	AUX	Heat 300-400°C	ATR
RWGS	AUX	CO2	AD
Compressor Syngas Medium to High Pressure	AUX	Electricity	MeOH Reactor
Compressor H2 Low to Medium Pressure	AUX	Electricity	AWE
Compressor CGH2	AUX	Electricity	AWE
Fuel Boiler	AUX	Heat 800-1000°C	ATR

The meteorological inputs required to model renewable energy generation, specifically solar irradiance and wind speed, are derived from coordinate-specific data for Johannesburg, South Africa, obtained via the open-source platform Open-Meteo. Given the location's placement in the Southern Hemisphere, the temporal definitions of the seasons are naturally inverted relative to standard Northern Hemisphere conventions. Accordingly, the temporal modeling parameters within this study define the summer period as spanning from November to March, while the winter season is defined from June to August.

3.1.1 MILP Optimization Framework Architecture

The computational framework developed for this study builds upon an existing modeling architecture at FEV Europe GmbH, utilizing the open-source energy system modeling library oemof.solph. This library functions as a high-level abstraction layer operating over the Pyomo optimization environment. Within this framework, all system components are mathematically formulated using strict linear representations. The model incorporates foundational linear constraints, including nodal mass and energy flow balances, component-specific operational boundaries (e.g., converter efficiency relations, generic storage state-of-charge dynamics, and

power-to-heat ratios for combined heat and power units), and capacity investment optimization limits. Furthermore, optional case-specific constraints are explicitly embedded into the formulation. To maintain computational tractability and prevent the exponential scaling of simulation times typically associated with complex energy networks, the introduction of binary variables was deliberately omitted. Consequently, the optimization of the hybrid system is defined exclusively as a continuous linear programming (LP) problem.

To systematically construct the energy topology, the operational units within the linear model are classified into five fundamental categorical objects:

- **Sources:** Foundational generation nodes that explicitly introduce energy or mass flows into the system boundary.
- **Converters:** Transformation nodes that convert a predefined set of input streams into corresponding output streams, governed by strict, fixed linear conversion efficiencies.
- **Buses:** Centralized connective hubs that route flows between the various topological components, ensuring nodal balance throughout the network.
- **Sinks:** Terminal nodes that exclusively receive input flows, representing system demands, required loads, or boundary out-streams.
- **Storages:** Temporal buffering components capable of absorbing and retaining incoming mass or energy flows across multiple timesteps, sequentially releasing them back into the system to satisfy dynamic operational requirements.

Components	Input	Output	Type
AD	Biomass, heat 60°C	Biogas, wastewater, digestate	Converter
MeOH Storage	Methanol	-	Sink
WT	-	Electricity	Sources
PV	-	Electricity	Sources
BESS	Electricity	Electricity	Storage
AWE	Electricity, water	Hydrogen, O ₂ , heat 60°C	Converter
ATR	Methane, Heat 1000°C, O ₂ , steam	Hydrogen, CO, wastewater	Converter

CGH2	Ultra-high pressure hydrogen	High pressure hydrogen	Storage
BGSS	Medium pressure methane	Medium pressure methane	Storage
Generator	Medium pressure methane	Electricity, CO2	Converter
MeOH reactor	Hydrogen, CO, CO2, Electricity	Methanol	Converter
Distillery	Methanol	Methanol	Converter
Biogas Compressor	Low pressure biogas	Medium pressure biogas	Converter
Biogas Separator (CO2 Removal)	Biogas	Methane, CO2	Converter
Steam Generator	Water, heat 300-400°C	Steam	Converter
RWGS	Hydrogen, CO2, heat 200-300°C	CO, Steam	Converter
Compressor Syngas Medium to High Pressure	Electricity, medium pressure syngas	High pressure syngas	Converter
Compressor H2 Low to Medium Pressure	Electricity, low pressure hydrogen	Medium pressure hydrogen	Converter
Compressor H2 Medium to High Pressure	Electricity, medium pressure hydrogen	High pressure hydrogen	Converter
Compressor CGH2	Electricity, medium pressure hydrogen	Ultra-high pressure hydrogen	AWE
Fuel Boiler	Medium pressure methane	Heat 800-1000°C	Converter
Heat exchangers (4 stages)	Heat high	Heat lower	Converter

The proposed Mixed-Integer Linear Programming (MILP) optimization framework is designed to operate in two distinct modes: the co-optimization of system sizing and operational dispatch, and the optimization of dispatch under fixed capacity constraints. The co-optimization pathway

concurrently evaluates optimal component capacities (sizing variables) alongside their corresponding operational schedules (dispatch variables). By formulating the system as a comprehensive linear programming problem, this approach balances the maximization of revenue with the minimization of both capital expenditures (CAPEX) and operational expenditures (OPEX). Conversely, the dispatch-only configuration utilizes predefined component sizes. Its primary function is to compute the optimal operational strategy that maximizes revenue and minimizes OPEX under fixed capacity constraints.

Upon defining the model architecture and selecting the operational mode, the formulation is processed using the COIN-OR Branch and Cut (CBC) MILP solver. This solver evaluates the overarching objective function subject to the established linear constraints to systematically identify the global optimal solution.

Mathematically, the primary objective function is formulated to minimize the net system cost. This is calculated as the sum of the annualized CAPEX and OPEX, minus the projected revenue (defined as the product of the total volume of methanol produced and the assumed market price of methanol). Furthermore, a supplementary performance metric—the unit cost of methanol production—is derived by dividing the sum of the annualized CAPEX and OPEX by the total volume of methanol produced.

$$\min Z = \text{CAPEX}_{\text{annualized}} + \text{OPEX}_{\text{annual}} - (M_{\text{MeOH, annual}} \cdot C_{\text{MeOH, assumed}})$$

Equation 15

$$\text{LCOM} = \frac{\text{CAPEX}_{\text{annualized}} + \text{OPEX}_{\text{annual}}}{M_{\text{MeOH, annual}}}$$

Equation 16

3.1.2 Reinforcement Learning Optimization Architecture

The RL optimization framework has also been established in-house at FEV Europe GmbH based on open source tools such as stable-baseline3, OpenAI gymnasium, pytorch as an alternative optimization solution introducing higher fidelity of per-component non-linear modelling and also

the novelty of reinforcement learning agent to optimize the operational dispatch of the system. The RL framework allows the adoption of both linear and non-linear model since the model is essentially a black box to the agent itself, thus, offering higher degree of freedom.

In the linear model, level 2 input, output, and conversion factors definition is fully utilized in order to have a 1-to-1 comparison with the MILP framework. By aligning all the techno-economic parameters, translating a global linear constraint of MILP into a step function-based RL framework, a fair comparison of these two methodologies can be made possible. Only then, it is also interesting to observe high-fidelity operation under different control strategies.

In the non-linear model, a non-linear high-fidelity physics modelling is taken into account for each component to achieve great physical accuracy. A series of calculation of per component step function evaluates the stream's journey from upstream to downstream processes. The table below summarizes the per component modelling of the non-linear model

Both the linear and non-linear model is built with the basis of a step function for each of the component's module. This step function is regulated by the evaluation order that dictates which component's step function is conducted first until last which is highly contrast compared to the MILP framework that encapsulates everything to one large equation. The evaluation order is highly integral in the behavior and the performance of the energy system model. Take the methanol RWGS and ATR as an example, if the step function for the RWGS occurs first before the compressor, the medium pressure of syngas will be empty when the RWGS request for the species, which in the end will result in infeasibility due to the deficit of the stream. Therefore, it is paramount to organize the evaluation order in a way that reflects the real operation of the plant. For this hybrid system, the evaluation order is summarized in the table below.

Components	Equation	Eval. order
AD		4
MeOH Storage		16
WT		2
PV		1

BESS	21
AWE	3
ATR	7
CGH2	14
BGSS	22
Generator	20
MeOH reactor	12
Distillery	15
Biogas Compressor	5
CO2 Removal	6
Steam Generator	17
RWGS	10
Compressor Syngas Medium to High Pressure	11
Compressor H2 Low to Medium Pressure	8
Compressor H2 Medium to High Pressure	9
Compressor CGH2	13
Fuel Boiler	19
Heat Exchangers	18

There are three available controllers in this framework: Conventional generalized rule-based controller (RBC), imposed control (IC), and RL. The RBC is a general rule that forces a component to simply process every stream passing through it and transferring the responsibility to satisfy remaining imbalances to a connected storage either to store or satisfy this imbalances. The IC controller is implemented as a validation step controller on running an equivalent linear model of the MILP framework in the RL framework before passing it to the RL controller. This controller simply takes the key components timeseries profile from the MILP framework, normalize them, and use them as an imposed controller for these key components, while the other components use

the RBC controller. If the result of this IC controller is equal to the MILP counterpart, only then it is clear that the linear model is 1-to-1 equivalent and ready to be run with the RL controller.

SAC algorithm has been selected for the RL controller due to its superiority in handling continuous dispatch with efficient use of data, and clear exploitation and exploration definition as learned in chapter 2. The figure below visualizes how the logic diagram of how the actor and critic agent optimizes the hybrid system.

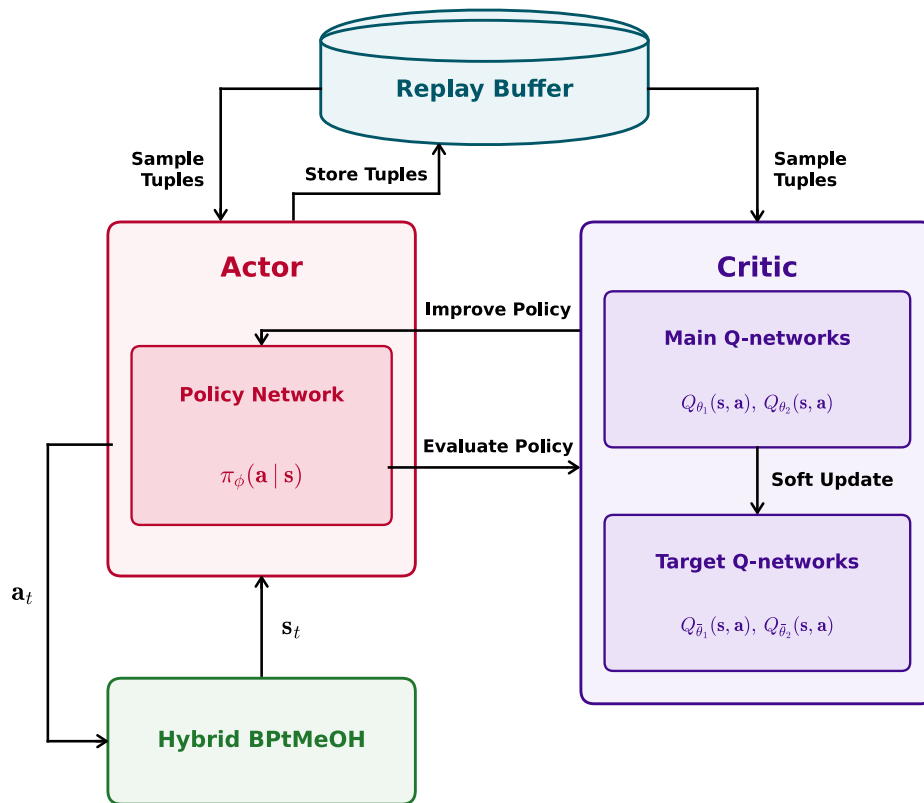


Figure 3 RL SAC algorithm process diagram [63].

The agent optimizes the hybrid system with the basis of an incentive defined in the reward function.

- Equation of the reward function
- The weight of reward and penalty of the reward function (Table)
- Agent hyperparameter

3.1.3 Bi-level Sizing and Dispatch Optimization Strategy

To further explore the significance of linear versus non-linear modeling in optimizing a complex hybrid energy system, a bi-level Bayesian Optimization (BO) strategy has been integrated with the existing framework. Adopting this architecture is a fundamental step in the methodology analysis for two primary reasons:

- **Economies of Scale (Non-linear CAPEX):** The inner-loop dispatch model has been intentionally formulated as a pure Linear Programming (LP) system to preserve computational efficiency, deliberately avoiding binary variables and piecewise linearization. Consequently, this strictly linear environment cannot natively model non-linear "economies of scale" for component capital expenditures (CAPEX). The bi-level strategy resolves this by preserving the rapid LP formulation for operational dispatch while wrapping a BO algorithm around it as an outer loop. In practice, the BO explores different equipment capacities and passes these as fixed boundaries to the inner LP loop. This iterative process allows the framework to accurately evaluate non-linear investment costs alongside operational dispatch, continuing until the BO locates the optimal combination of sizing capacities.

$$c = c_{ref} \cdot \left(\frac{S}{S_{ref}} \right)^k$$

Equation 17

- **Non-linear Objective Function (LCOM):** Beyond equipment scaling, a standard LP framework is mathematically restricted to linear objective functions, such as minimizing total absolute cost. However, the true optimal configuration for a hybrid methanol (MeOH) plant is dictated by the lowest Levelized Cost of Methanol (LCOM), rather than strictly the lowest overall expenditure. Because LCOM is a non-linear ratio, it cannot be optimized directly within the LP environment. The bi-level approach effectively shifts this non-linear evaluation outside the LP realm. For each iteration, the BO outer loop retrieves the total operating cost and the total MeOH produced from the inner loop. It then calculates the LCOM and uses this non-linear metric as its primary objective, guiding its probabilistic search—balancing exploration and exploitation—to pinpoint the global minimum.

As introduced in Chapter 2, Bayesian Optimization (BO) is a probabilistic, model-based

optimization technique that efficiently navigates complex search spaces by balancing the exploration of highly uncertain regions with the exploitation of known, high-performing areas. In this bi-level framework, the objective variable is defined as the Levelized Cost of Methanol (LCOM). The BO serves as an iterative outer loop, systematically proposing different combinations of component sizes and learning which configurations minimize the LCOM while ensuring the system remains physically viable. To enforce strict operational boundaries, the BO incorporates a penalty mechanism: if a proposed sizing configuration results in an infeasible dispatch within the inner LP loop (e.g., a bus imbalance or stream overflow), the algorithm assigns a severe mathematical penalty to that trial. This teaches the surrogate model to steer clear of similar unviable regions in future iterations. The figure below illustrates this iterative mapping process, demonstrating how BO systematically narrows down the optimal design space by strategically balancing exploration and exploitation.

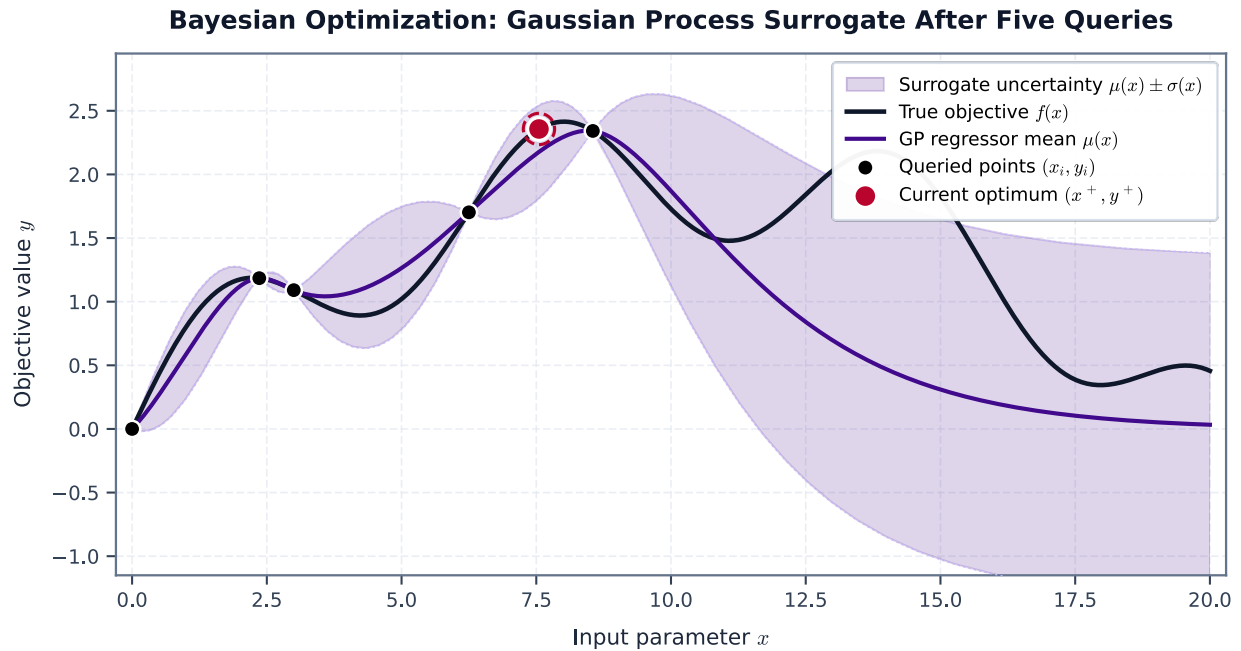


Figure 4 Gaussian process of bayesian optimization [64].

This figure illustrates the fundamental concept of Bayesian Optimization (BO) using a Gaussian Process (GP) after five queries or trials. The black line represents the true objective function that we are trying to map, and the purple line is the GP regressor mean, representing the surrogate

function estimated from the sampled data. By initializing a few trials to gain a sense of the true objective function, the GP calculates the probability distribution of the surrounding area. It then determines the next sampling point by balancing exploitation (sampling where the predicted cost is low) and exploration (sampling where uncertainty is high) through an acquisition function. A common acquisition function for this GP approach is the Lower Confidence Bound (LCB), defined as:

$$a_{\text{LCB}}(x) = \mu(x) - \kappa\sigma(x)$$

Equation 18

Where x is the specific plant design being evaluated, $\mu(x)$ is the predicted mean or the model's best guess of the cost, $\sigma(x)$ is the standard deviation representing the uncertainty at that point, and κ is the exploration parameter. By subtracting the scaled uncertainty from the predicted cost, the formula actively scores highly uncertain areas as potentially cheaper, encouraging the algorithm to explore them.

While Gaussian Processes provide a classic and intuitive visual representation of how Bayesian Optimization maps a search space, BO is a broader strategy rather than a single mathematical formulation; any algorithm that uses past results to update a probabilistic model and strategically select the next evaluation point qualifies as BO. In the bi-level optimization approach developed in this thesis, an open-source BO framework, Optuna, was utilized as the outer-loop layer to smartly select various sizing combinations for the 10 decision variables.

Rather than a standard GP, Optuna defaults to a highly efficient BO technique known as the Tree-structured Parzen Estimator (TPE). Unlike a GP, which attempts to predict the exact cost and uncertainty for every possible configuration, TPE models the search space by constructing two distinct probability density functions: one for the best-performing designs (e.g., the top 10% of trials) and one for the underperforming designs. Instead of using LCB, TPE mathematically selects the next trial by maximizing the probability that a new design belongs to the "good" distribution rather than the "bad" distribution. This approach inherently optimizes for Expected Improvement (EI). By utilizing Optuna's TPE sampler, the outer loop efficiently navigated the complex, 10-variable search space to find the optimal system sizing and minimize the Levelized Cost of Methanol (LCOM) without the heavy computational scaling limitations of a traditional GP.

Table 4 BO Hyperparameters

BO Hyperparameters	Values
Sampler	Tree-structured Parzen Estimator (TPE)
Number of trials	300
Number of startup trials	40
Number of jobs (parallelization)	4
Penalty cost	10^{20}
Sampler seed	42

3.2 Comparative Analysis Methodology for The Two Frameworks

This study employs a two-step comparative methodology to systematically evaluate the sizing and operational dispatch frameworks of a BPtMeOH system. The overarching goal is to determine the most robust optimization strategies by sequentially isolating structural sizing decisions from dynamic operational controls.

3.2.1 Phase 1: Joint Operational Dispatch and Sizing Methodology Comparison

The first phase evaluates joint operational dispatch and sizing methodologies through a nested optimization structure. The inner-loop operational decision-making is standardized utilizing an *oemof*-based formulation. The comparative analysis occurs primarily in the outer loop, which contrasts a co-optimization approach utilizing a Mixed-Integer Linear Programming (MILP) Investment Block (IB) against a bilevel formulation employing Bayesian Optimization (BO). The primary objective of this outer-loop comparison is to identify the optimal system layout under both linear and non-linear economies of scale, alongside other systemic parameters. Because the full MILP framework strictly optimizes for total cost and cannot inherently optimize for a non-linear Levelized Cost of Methanol (LCOM), a sensitivity analysis will be conducted for the MILP model. This will involve systematically varying initial methanol price assumptions to approximate non-linear cost dynamics.

To fully capture how the two co-optimization strategy optimal layouts behave under varying conditions, these three case study parameters will be applied to stress-test the two optimal layouts

as fixed capacities that was simulated on a baseline period of 2025. The scenarios encompass variations across three dimensions:

- **Economic Assumptions:** Comparing linear versus non-linear economies of scale.
- **Weather Variability:** Contrasting periods of stable solar and wind input against periods of extreme weather volatility that is based on the actual yearly average and standard deviation of the weather data on the chosen location for the past 10 years. 2018 has been chosen to represent stable weather whereas 2014 to represent the poor weather.
- **Market Conditions:** Evaluating current baseline parameters against forecasted 2030 market conditions.

The performance of the models across these stages and case studies will be evaluated using a predetermined set of Key Performance Indicators (KPIs). At the techno-economic level, the assessment will track LCOM, annualized Capital Expenditures (CAPEX) and Operational Expenditures (OPEX), carbon intensity, and clipped energy. At the framework level, the computational efficiency of each approach will be quantified by measuring total simulation time.

3.2.2 Phase 2: Operational Dispatch Optimization Methodology Comparison

The second phase of the methodology shifts the focus to comparing operational dispatch optimization frameworks, specifically evaluating MILP against Reinforcement Learning (RL). This comparison is conducted in two progressive stages, utilizing the fixed, optimal linear layout derived from the Phase 1 methodology.

- **Stage 1: Linear Model Reconstruction and Validation:** The first stage establishes a baselining and validation environment. A linear model reconstruction is performed within an RL step-function-based environment utilizing imposed controls. Based on findings from this reconstruction, RL control will be attached to five critical system components. Identifying and restricting these specific components is necessary to guarantee a strict one-to-one validation of the system design between the MILP and RL frameworks. Once a 1-to-1 system has been proven, the RL agent will be trained for this fixed linear model definition and the result will be compared with the MILP counterpart by observing the amount of methanol being produced during the simulation period by both approaches since CAPEX and OPEX of the system is identical between the two.
- **Stage 2: Non-Linear Model Integration with RL:** Following successful linear validation and

comparison, the methodology transitions to a non-linear modeling paradigm. The imposed control strategies established in Stage 1 will first be operated within a high-fidelity, non-linear system representation of the BPtMeOH process. Subsequently, an RL agent will be trained to autonomously manage this high-fidelity non-linear system. The final step involves a comparative analysis of the operational dispatch results, contrasting the performance of the high-fidelity BPtMeOH system under pre-calculated imposed controls intended for linear system against the autonomous RL framework trained specifically for the high-fidelity model.

4 Results and Discussions

4.1 Co-optimization MILP vs Bilevel Bayesian & MILP for optimal sizing and dispatch

- The results of both in 2025 in plot and table (same as presentation), discuss about the KPIs
- Case study analysis for capex assumption, weather sensitivity, forecast 2030 (Redo the co-optimization for all the sensitivity analysis)
 - For both MILP and BO results in 2025, run again in the different case study to show robustness in terms of how good do the layout still work under different weather and price assumption and how does the two methodology compared to each other
- BO trial analysis
- Short conclusion from this section

4.1.1 Sensitivity analysis for co-optimization MILP

- Show results under varying price methanol assumption since MILP framework could only optimize the total cost: Time convergence, layout, LCOM
- Explain that in the end the combination of BO and MILP rerun would produce a layout with the least LCOM or simply iterative MILP rerun using the first LCOM result as the next LCOM assumption

Figures & Tables:

- KPI plots
- Sizing tables
- BO convergence analysis plot
- CAPEX contribution plot for the forecast 2030 section

4.2 MILP vs Reinforcement learning for optimal dispatch

- Intro to the section

4.2.1 Linear system

- Linear system rebuilt as step function based system in the RL framework
 - Key components that require command in order to reach 1:1 behavior
 - Digester, AWE, ATR, RWGS, Generator, Curtailed
 - Learning points of linear system reconstruction

- Either battery or generator has to be controlled, it does not just operate with a RBC that takes in what is left because this logic is sometimes not optimal
 - Curtailed electricity has to be controlled in order to have a 1:1 behavior between MILP and linear RL
 - Evaluation order has to be reorganized after this findings
 - We used these key components as action spaces for RL to control and optimize
- RL system run on the linear system
 - Explanation and figure on how RL learns
 - Compare the amount of methanol produced
 - Compare the operational dispatch of key components (SOC, AWE)
 - Explain of the uniformity behavior of the neural network? And how to improve this and the result of the improvement attempt
 - Closing remarks
- RL agent run on different layout, can the RL agent still operate optimally?

Figures & Tables:

- Validation plot that shows linear system in MILP and RL framework is 1:1 (plot of key components): Include MeOH
- Plot of how the MILP linear system decides to curtail even before battery SOC reaches 100% and how this has to be reflected in RL system by reorganizing the evaluation order
- Comparison action plots between MILP behavior and RL result behavior
- Plot that shows RL uniformity behavior that can be linked to how the neural network operates
- Plot of RL learning progression based on rewards

4.2.2 High-fidelity non-linear system

- RL system run on the non-linear system using the linear trained agent
- RL system run on the non-linear system (train + eval)
 - Explanation and figure on how RL learns
 - Compare the amount of methanol produced
 - Compare the operational dispatch of key components (SOC, AWE)
 - Closing remarks

- Hypothesis:
 - The linear system is too overoptimistic, non-linear system produces less methanol, the linear trained agent will definitely perform poorly and less than the linear result one
 - The non-linear trained agent also produces less methanol compared to the linear one, but it is more realistic because the linear system oversimplifies the operation assumption such as the efficiency curve

5 Conclusions

Future outlook:

Stage 3: Both outerloop & Innerloop BO RL Nonlinear vs pathway [MI1.1]

- How different is the optimal results of a simplified linear model and accurate complex physics-based model?[]
- Future outlook: Bilevel with non-linear model with RL and BO

6 Annex

6.1.1 Data

- Economical Parameters definition (joint table): lead variable, capex, opex, reference scale, Scaling factor, reference

In this section, the data used for the hybrid MeOH model is presented along with the references. The table below summarizes the economic parameters of each component included and their references.

- Table from FEV consulting

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